Chain-of-Thought Reliability Test & Directional-Consistency Fix

sep_agent.py - 5 held-out patients, one per ACDC pathology group, use_rag=True throughout

1. Setup

- 5 test patients, one per ACDC pathology group (DCM, HCM, MINF, NOR, RV), all unseen by YOLO/U-Net/classifier training
- Same fixed real ECG record (00001_hr) paired with each, to isolate the MRI-side variation
- Each case: generate_sep_agent_decision run twice on the identical inputs - use_cot=False, then use_cot=True (both use_rag=True)
- Goal: does CoT change the output? Does it parse reliably? And - the actual question that matters - is the resulting text clinically consistent?

2. Original test - structural results

- Summary changed by CoT: 5/5
- First-attempt parse success (no CoT): 5/5
- First-attempt parse success (CoT): 5/5
- Reasoning field present when required: 5/5
- Mechanically, CoT is fully reliable - never crashes, never fails to parse, always changes the output

3. But reading the text: 2/5 real characterization errors

- Traceability guarantees a cited (agent_id, feature) pair is real - it does NOT guarantee the model's qualitative word for that value ("low"/"high"/"reduced"/"elevated") is the correct direction
- This is the same failure class already documented in sep_agent_demo.ipynb (a 295.5 mL LV volume once called "low" instead of elevated/dilated) - still present here, even with RAG's correctness_rule active

patient102 (true NOR, LV EF 70.9%, flag = high_lv_ejection_fraction_suspected)

> CoT reasoning: “...a decreased ejection fraction (70.9%)...”
> CoT summary: “The patient has an estimated heart rate of 63.9 bpm with a slightly elevated heart rate standard deviation. The left ventricle is enlarged and ejection fraction reduced, raising suspicion for dilated cardiomyopathy. Right ventricular volumes are also increased but there is no diagnostic evidence of abnormal RV function. There's some uncertainty regarding the presence of DCM.”

- 70.9% is elevated - it is exactly what triggered the high-EF flag, not a low-EF one
- The no-CoT summary for this same case avoided the error entirely by staying purely numeric (no qualitative adjective)

patient109 (true RV, RV EF 33.7%, flag = low_rv_ejection_fraction_suspected)

> CoT summary: “The patient exhibits a normal left ventricular ejection fraction of 52.4% with a mildly reduced right ventricular ejection fraction at 33.7%. A diagnosis of dilated cardiomyopathy is considered less likely (NOR probability: 99.7%) while hypertrophic cardiomyopathy and prior myocardial infarction are less likely. The ECG findings indicate a normal heart rate, standard deviation, and QRS duration. The patient has normal cardiac function with no signs of reduced ejection fraction.”

- Reasoning text itself was accurate (“...the right ventricle has a low ejection fraction (33.7%)”)
- But the final summary understates it (“mildly reduced”) then, one sentence later, flatly contradicts itself: “no signs of reduced ejection fraction”

- Error introduced only during the reasoning-to-summary compression step, not in the reasoning itself

Root cause common to both

- The correct direction was already known - it's the flag itself (high_lv_.../low_rv_..._suspected), computed deterministically in decide_next_steps_mri
- The model didn't need to judge the number's direction at all, and got it wrong anyway when it tried to

4. The fix

- Same pattern already used once in agent_common.py (MedGemma's image over-citation was fixed by force-correcting evidence_ref in code, not by prompting harder alone)
- New: _check_directional_consistency(text, flags_for_review) in sep_agent.py - a sentence-level keyword heuristic (not NLI/a formal proof)
- For each flag with a known direction (the 2 EF flags, tachycardia, bradycardia), scans reasoning + summary text for a sentence mentioning the relevant measurement (and ventricle, with 1-sentence lookback for anaphoric follow-ups like “...no signs of reduced ejection fraction”) alongside a word implying the OPPOSITE direction
- A violation is treated exactly like a fabricated citation: it raises SepAgentParseError, which re-enters the existing retry loop (no new retry infrastructure needed)
- Also added a fixed, unconditional prompt rule (not gated behind use_rag/use_cot): flags already indicate the correct direction, describe it consistently rather than independently judging the number
- Limitation, stated plainly: this is a keyword heuristic scoped only to the flags explicitly listed in the code - it can miss phrasing it doesn't recognize, and is not a general correctness prover

5. Re-run on the same 5 cases, after the fix

- Summary still changed by CoT: 5/5 (fix does not suppress CoT's actual effect on output)
- Succeeded within the retry budget: 5/5 (no CoT), 5/5 (CoT)
- Of those, a retry was triggered and then succeeded: 1/5 (no CoT), 1/5 (CoT)
- Final accepted output free of directional violations: 5/5 (no CoT), 5/5 (CoT)
- Cases that exhausted the retry budget without the model ever self-correcting: 0/5 - reported honestly below, not hidden

Before / after, the two known failure cases specifically

Patient	Flag	Before-fix CoT summary said...	After-fix CoT summary says...	Retry trigger
patient102	high_lv_ejection_fraction_suspected, pathology_NOR_pattern_suspected	The patient has an estimated heart rate of 63.9 bpm with a slightly elevated heart rate standard deviation. The left ventricle is enlarged and ejection fraction reduced, raising suspicion for dilated cardiomyopathy. Right ventricular volumes are also increased but there is no diagnostic evidence of abnormal RV function. There's some uncertainty regarding the presence of DCM.	The ECG shows a regular rhythm with an estimated mean heart rate of 63.9 bpm. The MRI reveals elevated LV volume (79.4 mL) and RV volume (49.8 mL), but also demonstrates a relatively preserved left ventricular ejection fraction (70.9%) and lower right ventricular ejection fraction (50.2%). The MRI agent flags dilated cardiomyopathy with a probability of 0.4. These findings suggest potential cardiac dysfunction, possibly related to dilated cardiomyopathy or other conditions affecting the heart chambers.	No

Patient	Flag	Before-fix CoT summary said...	After-fix CoT summary says...	Retry trigger
patient109	low_rv_ejection_fraction_suspected, pathology_NOR_pattern_suspected	The patient exhibits a normal left ventricular ejection fraction of 52.4% with a mildly reduced right ventricular ejection fraction at 33.7%. A diagnosis of dilated cardiomyopathy is considered less likely (NOR probability: 99.7%) while hypertrophic cardiomyopathy and prior myocardial infarction are less likely. The ECG findings indicate a normal heart rate, standard deviation, and QRS duration. The patient has normal cardiac function with no signs of reduced ejection fraction.	The patient exhibits normal sinus rhythm with a heart rate around 64 bpm, as indicated by the ECG. The left ventricle appears to be functioning normally with an ejection fraction of approximately 52%, and there's a suspicion of right ventricular dysfunction due to its volume and low ejection fraction.	Yes

6. Conclusion

- Structural CoT reliability was already solid (5/5) and remains so after the fix
- Directional-consistency violations in the final ACCEPTED output: 5/5 clean with CoT, 5/5 clean without CoT
- Both previously-broken cases (patient102, patient109) are covered by the new check - see table above for whether each specific case was corrected via retry
- This closes the loop opened by the sep_agent_demo.ipynb finding: traceability alone (citation is real) is necessary but not sufficient - a deterministic direction check was needed on top of it, same as evidence_ref needed a deterministic force-correction in agent_common.py
- Remaining honest limitation: the check only covers the flags explicitly listed in _DIRECTIONAL_FLAG_CHECKS (2 EF flags + tachycardia/bradycardia) - a similar characterization error on a flag not in that list (e.g. a pathology__pattern_suspected mischaracterization) would not be caught

7. Additional finding from this same re-run - not yet fixed

- patient104's CoT summary: “...The RV ejection fraction is slightly low at 65.6%.”
- 65.6% is not low - no low_rv_ejection_fraction_suspected flag fired for this patient (65.6% doesn't cross the 45% threshold), so the new directional check had no flag to check this claim against
- More importantly: this claim has NO traceability entry at all - none of the 8 entries in this response cite it
- This slips past every existing layer at once: traceability validation only checks that entries WHICH EXIST cite a real pair, not that every claim in the summary has one; the correctness_rule requiring a grounding citation is a prompt instruction only, not code-enforced; and the new directional check only fires for flags actually present
- Different bug category from the two fixed above (those were wrong-direction claims about an ALREADY-FLAGGED measurement) - this is an ungrounded, uncited claim about a measurement that never crossed any threshold
- Reported here honestly as a further, real gap found via the same test run, not folded into “fixed” - a proper fix would need a code-level check that every quantitative sentence in the summary has a matching traceability entry, which is future work, not implemented in this pass